

Talha Bilal

Senior Software Engineer & Technical Team Lead

Java | Spring Boot | PKI | CSC Remote Signing | Keycloak SPI | HSM

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Senior Java engineer & team lead - 5+ years in PKI, CSC remote signing, Keycloak (custom SPI), Java Card, and HSM platforms. Delivers production microservices for regulated, multi-tenant environments.

CORE SKILLS

Java 21/17, Spring Boot, Spring Security, REST / OpenAPI, Microservices, Event-driven (RabbitMQ), PKI / CA, OCSP / CRL, S/MIME & CMS, CSC API (remote signing), ETSI EN 319 401 / 411, PKCS#11, Keycloak, OAuth 2.0 / OIDC, Keycloak SPI (Authenticator, User Storage, Protocol Mapper), Bouncy Castle, IAIK, RFC 3161 TSA, Java Card 3.x, GlobalPlatform 2.x, APDU / host tools, AWS CloudHSM, Utimaco, SoftHSM2, Docker, CI/CD (Jenkins, GitHub Actions), PostgreSQL / MySQL, Redis, Integration & API testing, Structured logging & production support

EXPERIENCE

Dictalabs - Technical Team Lead - Java Spring Boot & PKI

Aug 2024 - Present | Enterprise digital trust & signing (B2B PKI platform)

- Lead engineers owning microservices for PKI, CMS/PKCS#7 signing, and HSM-backed key operations across production tenants.
- Architected PKCS#11 integration layer (AWS CloudHSM, SoftHSM2) - improved signing stability via pooled sessions, idempotent retries, and operational runbooks.
- Built CSC-compliant remote signing service (credentials, info, signatures, timestamps) with OAuth-secured API access and HSM-backed signature creation for client apps.
- Delivered end-to-end Keycloak: multi-realm OIDC, client policies, and custom SPIs (Authenticator, User Storage, Protocol Mapper) for certificate-aware login.
- Ship Java Card 3.x applets (RSA/ECDSA on-card) with GlobalPlatform lifecycle and host issuance tooling.
- Drive RFC 3161 timestamping, OCSP/CRL hardening, and incident response with security and product leadership.

Codegic - Senior Software Engineer - Java Spring Boot & PKI

Aug 2021 - Jul 2024 | PKI & secure document/signing products

- Built production CA stack: issuance, renewal, OCSP/CRL responders, and tenant-scoped certificate policies.
- Developed S/MIME/CMS email signing and Crypto-Core services integrated with enterprise HSMs and audit trails.
- Implemented Java Card applets and PC/SC host applications for smart-card issuance and secure key ceremonies.
- Established Docker-based CI/CD, API regression suites, and performance tuning for high-volume signing paths.
- Delivered multi-tenant PDF/signing workflows with approval chains and compliance-oriented logging.

SELECTED PROJECTS

- Keycloak IAM - OIDC realms, certificate-aware SPIs, multi-tenant auth
- CSC Remote Signing - credentials, signHash, OAuth, HSM-backed signatures
- PKI Server - CA, OCSP/CRL, HSM-backed multi-tenant issuance
- PKCS#11 Service - pooled HSM sessions, multi-vendor support
- Java Card - GP 2.x lifecycle, RSA/ECDSA applets, host integration
- Secure Email Signing - S/MIME/CMS, RSA & ECDSA
- Workflow Management - PDF signing, approvals, audit trails

HIGHLIGHTS

- Keycloak IAM with PKI-backed authentication: Single IAM layer for admin and customer apps, reduced bespoke auth code in services, and auditable login flows aligned with PKI issuance.
- CSC remote signing for client applications: Interoperable remote signing for web and mobile clients, reduced custom integration effort, and a single audited path from CSC requests to HSM-backed signatures.

EDUCATION

B.S. Computer Science - KFUEIT, 2021